1. Compute the gradient for the function $y=8x^4+ 3x^3 +7x^2+6x+3$ and verify the gradients provided by PyTorch with the analytical gradients.

2. For the following training data, build a linear regression model. Assume w and b are initialized with 1, and the learning rate is set to 0.001.
   x = torch.tensor([12.4, 14.3, 14.5, 14.9, 16.1, 16.9, 16.5, 15.4, 17.0, 17.9, 18.8, 20.3, 22.4, 19.4, 19.4, 15.7, 16.7, 17.3, 18.4, 19.4, 19.5, 19.7, 21.2])

   y = torch.tensor([11.2, 12.5, 12.7, 13.1, 14.1, 14.8, 14.4, 13.4, 14.9, 15.6, 16.4, 17.7, 19.6, 16.9, 14.0, 14.6, 15.1, 16.1, 16.8, 15.2, 17.0, 17.2, 18.6])

   Assume learning rate =0.001. Plot the graph of epoch on the x-axis and loss on the y-axis. Extend the code to use the California Housing Dataset

```
from sklearn.datasets import fetch_california_housing

# 1. Fetch the dataset

# Setting as_frame=True automatically returns it as a Pandas DataFrame

housing_data = fetch_california_housing(as_frame=True)
```

3. Implement logistic regression
   x = [1, 5, 10, 10, 25, 50, 70, 75, 100,]

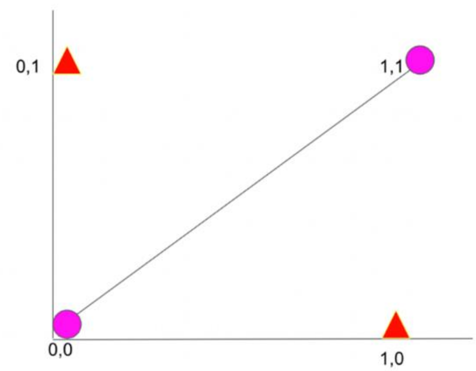
   y = [0, 0, 0, 0, 0, 1, 1, 1, 1]

   Extend the code to use the Titanic Dataset or the Breast Cancer Wisconsin (Diagnostic) Dataset.

```
#Reading the dataset

import seaborn as sns

# Load Titanic dataset for binary classification

titanic = sns.load_dataset('titanic')

print(titanic.head())
```

4. Implement two layer Feed Forward Neural Network for XOR Logic Gate with 2-bit Binary Input using Sigmoid activation. Verify the number of learnable parameters in the model.

X_1	X_2	$Y = X_1$ XOR X_2
0	0	0
0	1	1
1	0	1
1	1	0



5. Implement Feed Forward Neural Network with two hidden layers for classifying handwritten digits using the MNIST dataset. Display the classification accuracy in the form of a Confusion matrix. Verify the number of learnable parameters in the model.